PROSPERO & Literature
Paper-wise Comparison Report

Artificial Intelligence for Lung Ultrasound Diagnostic Accuracy

Prepared: 7 November 2026

Working document for research-team review; source-level verification remains necessary for items explicitly flagged below.

Scope: Review-level comparison, PROSPERO registration mapping, condition-wise primary-study comparison, cross-review overlap, and implications for the proposed four-condition diagnostic test accuracy (DTA) review.

Prepared: 7 November 2026

Review-level comparison

López-Canay et al. 2026

• **Scope:** Broad AI support for LUS interpretation, localization/detection of artifacts, anatomy and pathological findings

• **Search:** Web of Science, IEEE Xplore, PubMed; 2015–25 Nov 2025

• **Selection/sample:** 278 identified; 194 after dedup; 58 full text; 24 included

• **Included primary studies:** 24

• **AI methods:** Segmentation 10; object detection 4; saliency 10; CNNs

• **Synthesis:** No meta-analysis due heterogeneity of aims, data, target artifacts and metrics

• **Risk of bias:** PROBAST+AI; concerns mainly participants and analysis

• **Implication:** Broad AI-LUS overlap, but not a patient-level four-condition DTA meta-analysis; excludes classification-only and quantification-only tasks

• **Source:** https://doi.org/10.3389/fradi.2026.1899103

Kamal et al. 2026

• **Scope:** AI-assisted LUS pneumonia diagnosis across paediatric and adult populations

• **Search:** PubMed, Google Scholar, Web of Science, Scopus, Elsevier, ResearchGate, Cochrane; 2018–2025

• **Selection/sample:** 1,299 identified; 1,085 screened; 12 full text; 10 included

• **Included primary studies:** 10; 15,868 participants

• **AI methods:** ANN, CNN, hybrid, 3D video and segmentation systems

• **Synthesis:** Narrative synthesis; Se 88–93%, Sp 80–100%, accuracy 86–94.5%

• **Risk of bias:** QUADAS-2; six low overall risk, four some concerns

• **Implication:** Direct pneumonia overlap; cannot claim pneumonia is unreviewed

• **Source:** https://doi.org/10.1002/sono.70083

Lee & Wong Ching Hwai 2026

• **Scope:** AI diagnostic accuracy for pleural effusion on ultrasound

• **Search:** 84 records; comprehensive database search

• **Selection/sample:** 6 eligible; 2,951 patients, 58,392 images, 12,069 clips

• **Included primary studies:** 6

• **AI methods:** Bivariate model / sROC

• **Synthesis:** Pooled Se .92 (95% CI .85–.96), Sp .96 (.88–.99), AUC .98; I^2>99%

• **Risk of bias:** Detailed tool not available in accessible abstract

• **Implication:** PE meta-analysis already exists; do not claim first pooled PE estimate

• **Source:** https://doi.org/10.1002/jcu.70168

Marchi et al. 2026

• **Scope:** AI automated analysis of thoracic ultrasound for pleural effusion

• **Search:** MEDLINE, Scopus, Scholar, IEEE, CENTRAL, ClinicalTrials.gov; to 20 Aug 2025

• **Selection/sample:** 5 included; 7,565 patients

• **Included primary studies:** 5

• **AI methods:** CNNs: Reg-STN/CNN, ResNet, U-Net, EfficientNet-B0, Pleff-Net

• **Synthesis:** Narrative synthesis; no meta-analysis due heterogeneity; Se 70.6–100%, Sp 67–100%, AUC .77–.99

• **Risk of bias:** QUADAS-2 + GRADE; high risk patient selection/index test

• **Implication:** Direct PE overlap; external performance attenuation and dataset leakage concerns

• **Source:** https://doi.org/10.3390/diagnostics16010147

Kossoff et al. 2024

• **Scope:** Automated LUS analysis for pneumothorax diagnosis

• **Search:** MEDLINE, Embase, Web of Science, Scopus to 5 Jul 2024

• **Selection/sample:** 10 included; 8 retrospective, 2 prospective

• **Included primary studies:** 10

• **AI methods:** CNN predominant; rule-based iFAST and other classifiers

• **Synthesis:** No meta-analysis due heterogeneity

• **Risk of bias:** Modified QUADAS-2; patient selection high/unclear; index-test concerns when same data used for training/testing

• **Implication:** Direct pneumothorax overlap; distinguish lung-sliding feature classification from clinical PTX diagnosis

• **Source:** https://doi.org/10.7759/cureus.72896

Pneumonia: all 10 included primary studies

Correa et al. — 2018 — Automatic classification of pediatric pneumonia based on lung ultrasound pattern recognition

Population/N: Children <5 y; n=21

AI: ANN pattern recognition of B-line features

Sensitivity: 90.9%

Specificity: 100%

Accuracy: NR

Validation: Internal

Reference standard: Clinical diagnosis

Notes: Small proof-of-concept cohort

DOI: 10.1371/journal.pone.0206410

Born et al. — 2021 — Deep-learning based detection of COVID-19 using lung ultrasound imagery / POCOVID-Net

Population/N: Mixed COVID-19, bacterial pneumonia and healthy; n=216

AI: VGG16-based CNN; image/video classification

Sensitivity: 90%

Specificity: 96%

Accuracy: 90%

Validation: Internal/external assessment reported in source work

Reference standard: Expert/clinical labels

Notes: Heterogeneous source dataset; incomplete patient metadata

DOI: 10.3390/app11020672

Erfanian Ebadi et al. — 2021 — Automated detection of pneumonia in lung ultrasound using deep video classification for COVID-19

Population/N: Adults; n=300

AI: 3D CNN / video classification

Sensitivity: 90%

Specificity: 90%

Accuracy: 90%

Validation: Internal

Reference standard: RT-PCR plus clinical/radiological composite

Notes: Composite classes include consolidation/effusion; check class-specific 2x2 data

DOI: 10.1016/j.imu.2021.100687

Magrelli et al. — 2021 — Classification of lung disease in children by using lung ultrasound images and deep convolutional neural network

Population/N: 33 healthy + 22 bronchiolitis + 7 bacterial pneumonia; n=62

AI: VGG19, Xception, Inception-v3, Inception-ResNet-v2

Sensitivity: 91.5%*

Specificity: 95.86%*

Accuracy: 91.5%*

Validation: Internal; 4-fold CV

Reference standard: Clinical / expert paediatric diagnosis

Notes: *Review reports 3-class model figures; original paper reports separate tasks and model-specific metrics

DOI: 10.3389/fphys.2021.693448

Fang et al. — 2022 — Ultrasound image intelligent diagnosis in community-acquired pneumonia of children using CNN-based transfer learning

Population/N: Children 1 month–14 y; n=89

AI: AlexNet, VGG, Inception-v3, ResNet, DenseNet transfer learning

Sensitivity: 92%

Specificity: 80%

Accuracy: 87%

Validation: Internal; k-fold CV

Reference standard: Clinical diagnosis; compared with CXR/CT

Notes: 65 CAP and 24 non-CAP; single centre

DOI: 10.3389/fped.2022.1063587

Kuroda et al. — 2023 — AI-based point-of-care lung ultrasound for screening COVID-19 pneumonia: comparison with CT scans

Population/N: Adults; n=56

AI: AI-POCUS, 12-zone B-line assessment

Sensitivity: 92.3%

Specificity: 100%

Accuracy: 94.5%

Validation: CT comparison; novice operator

Reference standard: Chest CT

Notes: 12-zone patient-level metrics; zone-level sensitivity only 37.2%

DOI: 10.1371/journal.pone.0281127

Etter et al. — 2024 — Curated and annotated dataset of lung US images in Zambian children with clinical pneumonia

Population/N: Children 1–59 months; n=400

AI: Clinically annotated dataset / segmentation-driven AI

Sensitivity: 88–92%

Specificity: 80–100%

Accuracy: 93.7%

Validation: External validation reported in review

Reference standard: Clinical pneumonia based on WHO criteria

Notes: Full-text model evaluation and test-set N need separate verification

DOI: 10.1148/ryai.230147

Kessler et al. — 2024 — Development and testing of a deep learning algorithm to detect lung consolidation among children with pneumonia using hand-held ultrasound

Population/N: Children 18 months–17 y; n=107

AI: Hybrid CNN + MobileNet V3

Sensitivity: 88%

Specificity: 89%

Accuracy: 88.5%

Validation: Internal

Reference standard: Physician clinical diagnosis and ancillary findings

Notes: Video-level consolidation detection; 117 exams, 1,193 videos

DOI: 10.1371/journal.pone.0309109

Bolouri et al. — 2025 — AI enhances lung ultrasound interpretation across clinicians with varying expertise levels

Population/N: Adults; n=359

AI: U-Net / AI-assisted interpretation

Sensitivity: 89.1%

Specificity: 92.9%

Accuracy: NR

Validation: Internal; multi-reader multi-case

Reference standard: Expert LUS consensus

Notes: Study includes PE and consolidation/atelectasis; not a pure pneumonia cohort

DOI: 10.3390/diagnostics15172145

Nishio et al. — 2025 — Deep-learning classification and segmentation model for pneumonia

Population/N: Adults; n=14,258

AI: DL classification + segmentation

Sensitivity: NR

Specificity: NR

Accuracy: 86.57%

Validation: External

Reference standard: RT-PCR plus CT

Notes: Review table does not report sensitivity/specificity; exact title/DOI verify from reference list

DOI: Full citation verification pending

Pneumothorax: all 10 included primary studies

Summers et al. — 2016 — Automated/iFAST analysis for pneumothorax

Population/N: N NR

AI: Rule-based diagnostic algorithm

Sensitivity: 79%

Specificity: 87%

Accuracy: NR

Validation: Separate test dataset

Reference standard: Clinician consensus

Notes: Early retrospective ED study; B-mode 107 + M-mode 26 scans

DOI: Reference [16] in Kossoff 2024

Jaščur et al. — 2021 — Detecting absence of lung sliding in lung ultrasound using deep learning

Population/N: Post-thoracic surgery; n=48

AI: ResNet-18 CNN + U-Net tissue segmentation

Sensitivity: 82%

Specificity: 92%

Accuracy: 89%

Validation: Internal; same dataset train/test

Reference standard: Clinician labelling

Notes: Same 48-patient dataset as Kolarik 2023; non-independent cohort

DOI: 10.3390/app11156976

VanBerlo et al. — 2022 — Accurate assessment of lung sliding artefact using deep learning

Population/N: ED/ICU/internal medicine; n=738

AI: EfficientNetB0 CNN + automask/ROI detector

Sensitivity: 94%

Specificity: 87%

Accuracy: 89%

Validation: Internal; patient-level 85/15 split

Reference standard: Clinician labelling with review

Notes: Same classifier lineage later evaluated by Fiedler and Wu

DOI: 10.1016/j.compbiomed.2022.105953

Montgomery et al. — 2023 — Automated ultrasound classification for lung sliding / pneumothorax

Population/N: n=45

AI: 2D U-Net CNN + 3D dynamic U-Net

Sensitivity: 86%

Specificity: 75%

Accuracy: NR

Validation: Separate test dataset

Reference standard: Clinician consensus

Notes: Small cohort; setting NR

DOI: Reference [19] in Kossoff 2024

Zhang et al. — 2023 — AI analysis of lung sliding and lung point

Population/N: n=204

AI: EfficientNetB0 CNN + denoising CNN

Sensitivity: 98%

Specificity: 98%

Accuracy: NR

Validation: Internal; 22.5% test

Reference standard: Clinician labelling

Notes: Primary outcome averaged lung sliding and lung point metrics

DOI: Reference [20] in Kossoff 2024

Kolarik et al. — 2023 — Detecting absence of lung sliding in videos using 3D CNN

Population/N: Post-thoracic surgery; n=48

AI: ResNet3D-18

Sensitivity: 94%

Specificity: 79%

Accuracy: 92%

Validation: Internal; 30% test

Reference standard: Clinician labelling

Notes: Same cohort as Jaščur et al.

DOI: 10.12700/APH.20.6.2023.6.3

Kim et al. — 2023 — Deep learning classification of lung sliding using M-mode ultrasound

Population/N: ED patients; n=77

AI: EfficientNet-lite0 + quality assurance / ROI detection

Sensitivity: 88%

Specificity: 80%

Accuracy: 84%

Validation: Separate test dataset

Reference standard: Clinician consensus

Notes: Whole pipeline includes quality assurance

DOI: Reference [22] in Kossoff 2024

Yang et al. — 2024 — Prospective AI detection of pneumothorax using lung ultrasound

Population/N: Critical-care dyspnoea; n=75

AI: AI classifier; architecture NR

Sensitivity: 79%

Specificity: 85%

Accuracy: NR

Validation: Prospective; separate test dataset

Reference standard: Chest CT

Notes: One of two prospective studies; BLUE protocol

DOI: Reference [23] in Kossoff 2024

Fiedler et al. — 2024 — Testing AI lung-sliding classifier for pneumothorax

Population/N: Thoracic surgery ward/ICU; n=62

AI: EfficientNetB0; VanBerlo model lineage

Sensitivity: 92%

Specificity: 80%

Accuracy: 82%

Validation: Prospective; separate test dataset

Reference standard: Clinician consensus

Notes: Indeterminate clips excluded from primary outcome

DOI: Reference [24] in Kossoff 2024

Wu et al. — 2024 — External testing/fine-tuning of AI lung-sliding classifier

Population/N: ED/ICU, 3 hospitals; n=238

AI: EfficientNetB0; VanBerlo model lineage

Sensitivity: 92%

Specificity: 82%

Accuracy: 83%

Validation: External dataset; one-third fine-tuned

Reference standard: Clinician labelling with review

Notes: Model lineage shared with VanBerlo/Fiedler

DOI: Reference [25] in Kossoff 2024

Pleural effusion: all 5 included primary studies

Tsai et al. — 2021 — Automatic deep learning-based pleural effusion classification in lung ultrasound

Population/N: n=70; 99,209 images from 623 videos

AI: CNN + regularised spatial transformer network

Sensitivity: NR

Specificity: NR

Accuracy: Frame accuracy 92.38%; video 91.12%

Validation: Internal; 10-fold CV

Reference standard: Expert clinician interpretation

Notes: F1/precision/recall reported; Se/Sp not individually reported

DOI: 10.1016/j.ejmp.2021.03.027

Chen et al. — 2023 — Intelligent interpretation of four lung ultrasonographic features with split attention DL

Population/N: n=3,966; 5,545 images

AI: ResNet + Split-Attention

Sensitivity: 96.39% PE class

Specificity: 100% PE class

Accuracy: 96.39% PE class

Validation: Internal; 80/20 image split

Reference standard: Two experienced ultrasonographers

Notes: Image-level multiclass; overall Se 98.27%, Sp 99.41%, Acc 98.20%, macro-AUC 99.76%

DOI: 10.1016/j.bspc.2022.104228

Huang et al. — 2024 — AI pleural effusion detection and segmentation using U-Net architectures

Population/N: n=848; 1,440 images

AI: Attention U-Net and U-Net

Sensitivity: 97% / 97%

Specificity: 91% / 86%

Accuracy: 94% / 92%

Validation: Internal; 10-fold CV

Reference standard: Manual annotation by two experienced sonographers

Notes: AUC 0.98 / 0.97; Dice 0.86 / 0.82; ranges reported

DOI: DOI verification pending

Hong et al. — 2024 — Multicentre deep learning classification of LUS findings including pleural effusion

Population/N: n=1,898 total

AI: EfficientNet-B0 transfer learning

Sensitivity: Internal 86.9%; ext1 85.5%; ext2 70.6%

Specificity: Internal 85.6%; ext1 76.5%; ext2 86.1%

Accuracy: Temporal test 86.2%

Validation: Internal, temporal and two external tests

Reference standard: Expert radiologist; CT in observer subset

Notes: AUC 0.94 internal/temporal, 0.89 ext1, 0.77 ext2

DOI: DOI verification pending

Chaudhary et al. — 2025 — Diagnostic accuracy of automated classifier for pleural effusion on LUS

Population/N: n=785; 1,664 videos; 313,109 frames

AI: Pleff-Net CNN; frame-to-clip aggregation

Sensitivity: General 90.3%; large effusion 95.5%; trauma model 98%

Specificity: General 89%; trauma model 67%

Accuracy: General 89.5%

Validation: Internal patient-level holdout / 10-fold CV

Reference standard: Expert LUS interpretation

Notes: AUC 0.939; 85/15 patient-level split; multi-centre

DOI: 10.1016/j.ajem.2025.01.041

Lee 2026: six-study mapping

Chaudhary et al. — 2025 — Diagnostic accuracy of automated classifier for pleural effusion

N: 785

AI: Pleff-Net CNN

Se: 90.3%

Sp: 89.0%

Acc: 89.5%

AUC: 0.939

Reference standard: Expert LUS interpretation

Notes: Also in Marchi 2026 and Frontiers 2026; cohort overlap check

Chen et al. — 2023 — Intelligent interpretation of four lung ultrasonographic features

N: 3,966

AI: ResNet + Split-Attention

Se: 96.39% PE class

Sp: 100% PE class

Acc: 96.39% PE class

AUC: 99.80% PE class

Reference standard: Expert ultrasonographers

Notes: Also in Marchi and Frontiers; image-level classification

Sagreiya et al. — 2023 — Automated lung ultrasound pulmonary disease quantification using unsupervised ML for COVID-19

N: NR

AI: Unsupervised ML

Se: NR

Sp: NR

Acc: NR

AUC: NR

Reference standard: Not verified from accessible article table

Notes: Appears among Lee article references; exact eligibility contribution needs full-text check

Shea et al. — 2023 — Deep learning video classification of LUS features associated with pneumonia

N: 762 patients; 22,400 scans/videos

AI: Deep-learning video classification

Se: NR

Sp: NR

Acc: NR

AUC: AUC ~0.90 for feature tasks

Reference standard: Expert/clinical annotation

Notes: Multi-condition features include PE; not necessarily patient-level PE DTA

Tsai et al. — 2021 — Automatic deep learning-based pleural effusion classification in LUS

N: 70

AI: CNN + Reg-STN

Se: NR

Sp: NR

Acc: Frame acc 92.38%; video 91.12%

AUC: NR

Reference standard: Expert LUS interpretation

Notes: Also in Marchi; frame/video metrics not pooled Se/Sp

Vukovic et al. — 2023 — Automatic deep learning-based pleural effusion segmentation in LUS

N: 24

AI: STN + U-Net

Se: NR

Sp: NR

Acc: NR

AUC: NR

Reference standard: Expert LUS annotation

Notes: Also in Frontiers; segmentation DSC 0.70, not clinical DTA

Frontiers 2026: all 24 included studies

Howell et al. — 2024 — Deep learning for real-time multi-class segmentation of artefacts in LUS

Population: UK; 41 COVID-19 pneumonia patients

Sample: 57 images

AI: Lightweight U-Net

Target: Pleural line, consolidation, simple/complex effusion

Validation: Internal; 51 train / 6 test

Metrics: DSC 0.20; IoU NR

DTA relevance: Not patient-level DTA

DOI: 10.1016/j.ultras.2024.107251

Frank et al. — 2022 — Integrating domain knowledge into deep networks for LUS with COVID-19 applications

Population: Israel / ICLUS, Italy; 33 patients + controls

Sample: 2,154 images

AI: DeepLabV3++ / ResNet-50

Target: A-lines, B-lines, consolidation severity, pleural line

Validation: Internal; patient-level 74/26

Metrics: DSC 0.70

DTA relevance: Segmentation/ordinal task, not clinical DTA

DOI: 10.1109/TMI.2021.3117246

Muñoz et al. — 2024 — Deep learning algorithms for real-time LUS assisted diagnosis

Population: Spain; 30 patients

Sample: 689 videos

AI: Attention U-Net

Target: Pleura, A-lines, B-lines, consolidation

Validation: Internal; patient-level training/validation

Metrics: DSC: pleura .83, A-lines .40, B-lines .87, consolidation .96

DTA relevance: Segmentation metrics, not clinical DTA

DOI: 10.3390/app142411930

Uchida et al. — 2024 — Automatic detection of pleural line and lung sliding using CNNs

Population: Japan; population NR

Sample: 10,140 images

AI: U-Net

Target: Pleural line segmentation

Validation: Internal; 98.6/1.4 split

Metrics: DSC .99; IoU .98

DTA relevance: Not patient-level DTA

DOI: 10.1016/j.heliyon.2024.e34700

Abbasi et al. — 2025 — Improved A-line and B-line detection using boundary-aware dice loss

Population: COVID-19 patients; N NR

Sample: 4,599 images

AI: TransBound-UNet

Target: A-lines, B-lines

Validation: Internal; 5-fold CV

Metrics: DSC .80; IoU .73

DTA relevance: Not patient-level DTA

DOI: 10.3390/bioengineering12030311

Lucassen et al. — 2023 — Deep learning for detection/localization of B-lines

Population: USA ED; 113 patients

Sample: 719 videos; 10,755 images

AI: EfficientNet-18 + U-Net

Target: B-line localization

Validation: Internal; patient-level 80/20

Metrics: Segmentation/localization metrics NR

DTA relevance: Feature localization, not clinical DTA

DOI: 10.1109/JBHI.2023.3282596

Jaščur et al. — 2021 — Detecting absence of lung sliding using deep learning

Population: Slovakia; 48 post-thoracic surgery patients

Sample: 545 videos; 6,400 images

AI: U-Net

Target: Lung, pleural line, rib shadow

Validation: Internal 50/25/25

Metrics: IoU .75/.61/.81

DTA relevance: No direct DTA in Frontiers; Kossoff reports Se .82 Sp .92

DOI: 10.3390/app11156976

Vukovic et al. — 2023 — Automatic deep learning-based pleural effusion segmentation

Population: Australia; 24 PE patients

Sample: 51 videos; 3,041 images

AI: STN + U-Net

Target: Pleural effusion segmentation

Validation: Internal patient-level 67/12/21

Metrics: DSC .70

DTA relevance: Segmentation, not patient-level DTA

DOI: 10.1186/s12911-023-02362-6

Xing et al. — 2024 — Combining quantitative and qualitative analysis for scoring pleural line

Population: China; 31 COVID-19 patients

Sample: 740 videos; 3,770 images

AI: Cascaded encoder-decoder CNN/MLP

Target: Pleural line

Validation: External; patient-level split

Metrics: DSC .87

DTA relevance: Not patient-level DTA

DOI: 10.1088/1361-6560/ad3888

Tan et al. — 2022 — Automated LUS image assessment to identify fluid overload in dialysis patients

Population: Singapore; 76 dialysis patients

Sample: 1,385 images

AI: Mask R-CNN + YOLACT

Target: B-lines

Validation: Internal patient-level 80/20

Metrics: DSC/IoU NR

DTA relevance: B-line/fluid overload feature, not disease DTA

DOI: 10.1186/s12882-022-03044-7

Xing et al. — 2023 — Automatic detection of A-line using deep learning and image processing

Population: China; 61 patients incl. 30 pneumothorax

Sample: 4,930 images

AI: Faster R-CNN / ResNet-50

Target: Pleural-line bounding box

Validation: External; patient-level 64/36

Metrics: mAP/Se/precision NR

DTA relevance: Target feature detection, not disease DTA

DOI: 10.1002/mp.15908

Bottino et al. — 2025 — Automatic B-line detection using YOLO

Population: Italy; 46 patients

Sample: 386 images

AI: YOLO / CSPDarkNet53

Target: B-line bounding boxes

Validation: Internal 80/20 CV

Metrics: Sensitivity 81%; precision 92%

DTA relevance: Not patient-level DTA

DOI: 10.1007/s40477-025-01077-w

Tripathi et al. — 2023 — Unsupervised landmark detection and classification of lung infection using transporter neural networks

Population: India/Spain; 40 patients

Sample: 230 videos

AI: KeyNet

Target: Pleura, A-lines, B-lines landmarks

Validation: Internal

Metrics: Sensitivity 99%; precision 83%

DTA relevance: Landmark detection, not diagnostic sensitivity

DOI: 10.1016/j.compbiomed.2022.106345

Joseph et al. — 2023 — lungEcho: resource-constrained LUS video analysis for triaging

Population: India/Spain; 100 COVID-19 patients

Sample: 1,200 videos

AI: YOLOv5s

Target: Pleura, ribs, shadows, A/B-lines, consolidations

Validation: Internal 70/20/10

Metrics: mAP 66%; Se 68.5%; precision 67%

DTA relevance: Detection metrics not patient-level DTA

DOI: 10.1016/j.bea.2023.100094

van Sloun et al. — 2020 — Localizing B-lines by weakly supervised deep learning

Population: Netherlands; 10 patients

Sample: 12 videos; 4,218 images

AI: CNN + Grad-CAM

Target: B-line presence/absence

Validation: Internal 8 train / 4 CV

Metrics: AUC .874; Se 85.6%; Sp 69.7%

DTA relevance: Frame/video classification, not patient DTA

DOI: 10.1109/JBHI.2019.2936151

Chaudhary et al. — 2025 — Diagnostic accuracy of automated classifier for pleural effusions

Population: Canada; 785 patients

Sample: 1,664 videos; 313,109 images

AI: EfficientNetB0 CNN + Grad-CAM

Target: PE present/absent

Validation: Internal patient-level 85/15

Metrics: AUC .939; Se 85.9%; Sp 89.3%

DTA relevance: Direct PE accuracy study; overlaps Marchi/Lee

DOI: 10.1016/j.ajem.2025.01.041

VanBerlo et al. — 2022 — Accurate assessment of lung sliding artefact using deep learning

Population: Canada; 738 patients

Sample: 3,075 videos

AI: EfficientNetB0 CNN + Grad-CAM

Target: Lung sliding present/absent

Validation: Internal patient-level 85/15

Metrics: AUC .973; Se 93.5%; Sp 87.3%

DTA relevance: Feature classifier; related model lineage in pneumothorax review

DOI: 10.1016/j.compbiomed.2022.105953

Kolarik et al. — 2023 — Detecting absence of lung sliding in videos using 3D CNN

Population: Slovakia; 48 patients

Sample: 48 videos

AI: ResNet3D-18 + saliency

Target: Lung sliding

Validation: Internal patient-level 70/30

Metrics: Se 93.6%; Sp 78.53%

DTA relevance: Same 48-patient cohort as Jaščur; avoid double-counting

DOI: 10.12700/APH.20.6.2023.6.3

Ni et al. — 2024 — Active learning with multi-enhanced views for multiple LUS patterns

Population: China; 3,966 patients

Sample: 5,267 images

AI: MEVAL CNN / ResNet50 + Grad-CAM

Target: A/B-lines, consolidation, PE

Validation: Internal; patient-level CV/test

Metrics: AUC .9989; Se 98.76%; Sp 98.6%

DTA relevance: Image-level multiclass; same source cohort as Chen

DOI: 10.1016/j.compmedimag.2024.102454

Chen et al. — 2023 — Intelligent interpretation of four LUS features with split attention DL

Population: China; 3,966 patients

Sample: 5,545 images

AI: ResNet + Split-Attention + Grad-CAM

Target: A/B-lines, consolidation, PE

Validation: Internal 80/20 image split

Metrics: AUC .9976; Se 98.27%; Sp 99.41%

DTA relevance: Image-level multiclass; same cohort as Ni; overlaps PE reviews

DOI: 10.1016/j.bspc.2022.104228

Arntfield et al. — 2021 — Automation of LUS interpretation for normal vs abnormal parenchyma

Population: Canada; patient N NR

Sample: 1,276 videos

AI: VGG16 CNN + Grad-CAM

Target: A-lines/B-lines; normal/abnormal

Validation: External validation

Metrics: AUC .93; Se 83%; Sp 82%

DTA relevance: External validation; not target-condition DTA

DOI: 10.3390/diagnostics11112049

Huang et al. — 2022 — Evaluation of pulmonary edema in COVID-19 pneumonia using NCA-ResNet

Population: China; 31 COVID-19 patients

Sample: 2,062 images

AI: NCA-ResNet + Grad-CAM

Target: B-line patterns / subtypes

Validation: Internal patient-level 80/20

Metrics: Se 90.43%; Sp NR

DTA relevance: Image-level artifact classification

DOI: 10.1016/j.ultrasmedbio.2022.01.023

Pare et al. — 2024 — Transfer-learning B-line assessment for acute heart failure

Population: USA; 90 acute HF patients

Sample: 716 videos

AI: ResNet18 + Grad-CAM

Target: B-line severity classes

Validation: Internal

Metrics: AUC .967; Se 96.3%; Sp 92.4%

DTA relevance: Heart-failure/B-line feature outcome; adjacent to interstitial syndrome

DOI: 10.1016/j.ultrasmedbio.2024.02.004

Erfanian Ebadi et al. — 2021 — Automated pneumonia detection using deep video classification for COVID-19

Population: Canada; 300 patients

Sample: 1,530 videos; 287,549 images

AI: 3D CNN + saliency

Target: A-lines, B-lines, consolidation/effusion

Validation: Internal 80/20 CV

Metrics: AUC .94/.91/.96; Se 91/86/92 by feature

DTA relevance: Composite output; condition-specific pooling limited

DOI: 10.1016/j.imu.2021.100687

Cross-review overlap

Chaudhary et al. 2025

Reviews: Frontiers 2026; Marchi 2026; Lee 2026

Condition: Pleural effusion

Deduplication warning: Same 785-patient / 1,664-video study; deduplicate primary evidence

Confidence: High

Chen et al. 2023

Reviews: Frontiers 2026; Marchi 2026; Lee 2026

Condition: A-lines, B-lines, consolidation, PE

Deduplication warning: 3,966-patient Shanghai cohort; image-level multiclass

Confidence: High

Vukovic et al. 2023

Reviews: Frontiers 2026; Lee 2026

Condition: Pleural effusion

Deduplication warning: 24-patient segmentation cohort; not patient-level DTA

Confidence: High

Tsai et al. 2021

Reviews: Marchi 2026; Lee 2026

Condition: Pleural effusion

Deduplication warning: 70 patients; 99,209 images from 623 videos

Confidence: High

Jaščur et al. 2021

Reviews: Frontiers 2026; Kossoff 2024

Condition: Lung sliding / pneumothorax proxy

Deduplication warning: 48-patient postoperative cohort also used by Kolarik

Confidence: High

Kolarik et al. 2023

Reviews: Frontiers 2026; Kossoff 2024

Condition: Lung sliding / pneumothorax proxy

Deduplication warning: Same 48-patient cohort as Jaščur

Confidence: High

VanBerlo et al. 2022

Reviews: Frontiers 2026; Kossoff 2024

Condition: Lung sliding / pneumothorax proxy

Deduplication warning: 738-patient model lineage later evaluated by Fiedler/Wu

Confidence: High

Fiedler 2024; Wu 2024

Reviews: Kossoff 2024; model lineage linked to VanBerlo

Condition: Lung sliding / pneumothorax proxy

Deduplication warning: Same EfficientNetB0 classifier lineage; datasets differ but not fully independent model evidence

Confidence: Moderate

Erfanian Ebadi et al. 2021

Reviews: Kamal 2026; Frontiers 2026

Condition: COVID-19 pneumonia / LUS feature classes

Deduplication warning: Same primary study appears in both broad and pneumonia-specific review

Confidence: High

Magrelli et al. 2021

Reviews: Kamal 2026

Condition: Paediatric pneumonia / bronchiolitis

Deduplication warning: Small paediatric sample; exact metric task distinctions matter

Confidence: High

Overall interpretation

The broad AI-LUS topic is already systematically reviewed. Pneumonia, pneumothorax and pleural effusion also have condition-specific reviews, and pleural effusion already has a published bivariate DTA meta-analysis. Therefore, the proposed project must not claim that no prior review exists.

The defensible potential contribution is a common DTA framework across four target conditions, plus prespecified care-setting, age, income-setting and external-validation comparisons. This is conditional on enough independent primary studies reporting compatible patient-level 2×2 data.

Critical distinction: Frontiers' 24 studies include segmentation, localization, object detection and feature-classification papers. Metrics such as DSC, IoU, mAP and frame-level sensitivity are not interchangeable with patient-level diagnostic sensitivity/specificity. These papers must not all be pooled as clinical DTA studies.

Limitations: The Lee six-study list is mapped from the six matching AI-LUS primary research papers cited in the article and needs a final check against the full-text inclusion table. Some exact DOIs and table-level metrics in Kamal/Marchi are flagged for verification. Scopus was not completed. PROSPERO per-record status was not retained in the supplied TXT. IEEE 135 records are preliminary title/abstract screening only.